

Capstone Panel Presentation — Technical Speaking & Viva Guide

Cancelable Multimodal Biometric Authentication for Critical Infrastructure

1. Maanasvee Khetan

Datasets, training, preprocessing & feature extraction

2. Reya Oberoi

Cancelable biometric & template architecture

3. Sanya Malik

Authentication, matching, fusion & security response

Every value in this guide was checked against the current code on 25 September 2026. Where the requested presentation plan differed from the code, the code wins and the difference is marked **Correction**.

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Part 1 — Presentation at a glance

Order	Presenter	Main responsibility	Topics	Hands over with
1	Maanasvee Khetan	Datasets, training, preprocessing, feature extraction	datasets, sizes, splits · training (face, voice, fingerprint) · pretrained vs trained · face detection, landmarks, alignment status, quality gates · five-capture enrollment · voice and fingerprint preprocessing · embeddings (512 / 192 / 512) · evaluation protocol and dataset limits	"...Reya will show how that embedding is protected."
2	Reya Oberoi	Cancelable biometric and template architecture	why cancellable biometrics · BioHash · HKDF-SHA256 keys · projection, quantisation, permutation · 256-bit / 32-byte template · four template sets · ACTIVE / STANDBY / REVOKED · revocation · unlinkability · storage	"...Sanya will show how these templates decide a login."
3	Sanya Malik	Authentication, matching, fusion and security response	authentication flow · Hamming similarity → estimated metrics · face / voice / fingerprint decision rules · thresholds (configuration values) · fusion policies and normalisation · granted / denied · rotation · concurrency · audit · authentication results and limits	Closes with the full-system summary.

Approximate timing: about 4–5 minutes each, then questions. The story now follows the data through the system as one presentation: Maanasvee explains how a raw biometric becomes an embedding; Reya explains how that embedding becomes a protected, revocable template; Sanya explains how those templates decide a login and how the system responds to a failure.

Evidence labels used in this guide

Label	Meaning	Example
REAL DATA	measured on real biometric datasets (LFW, VoxCeleb1 subset, SOCOFing)	face protected EER 8.50 %
SYNTHETIC	measured on generated vectors (not people)	BioHash calibration curve, $R^2 \approx 0.9999$
SOFTWARE TEST	automated tests of the code — not biometric trials	657 backend tests passing
CONFIGURATION	a value set in the code or settings	face threshold 0.80, voice 0.75
DERIVED	computed from other values	$0.80 \approx$ at least 210 of 256 bits equal
FUTURE WORK	not done / not available	real-user study; data-selected thresholds

The whole system, with owners

USER INPUT (camera frame / voice WAV / fingerprint image)	[MAANASVEE]
↓	
PREPROCESSING (face detect + crop · audio clean-up · ridge enhance)	[MAANASVEE]
↓	
EMBEDDING (512 / 192 / 512 numbers, scaled to length 1)	[MAANASVEE]
↓	
CANCELLABLE TRANSFORMATION (HKDF key + BioHash)	[REYA]
↓	
256-BIT TEMPLATE (4 sets: T1 ACTIVE, T2–T4 STANDBY)	[REYA]
↓	
MATCHING (Hamming similarity → estimated metric)	[SANYA]
↓	
FUSION (ALL_REQUIRED: every presented factor must pass)	[SANYA]
↓	
ACCESS DECISION (GRANTED / DENIED)	[SANYA]
↓	
FAILED AUTHENTICATION (backend decision only)	[SANYA]
↓	
TEMPLATE ROTATION (Tn REVOKED → Tn+1 ACTIVE, if enabled)	[SANYA + REYA]

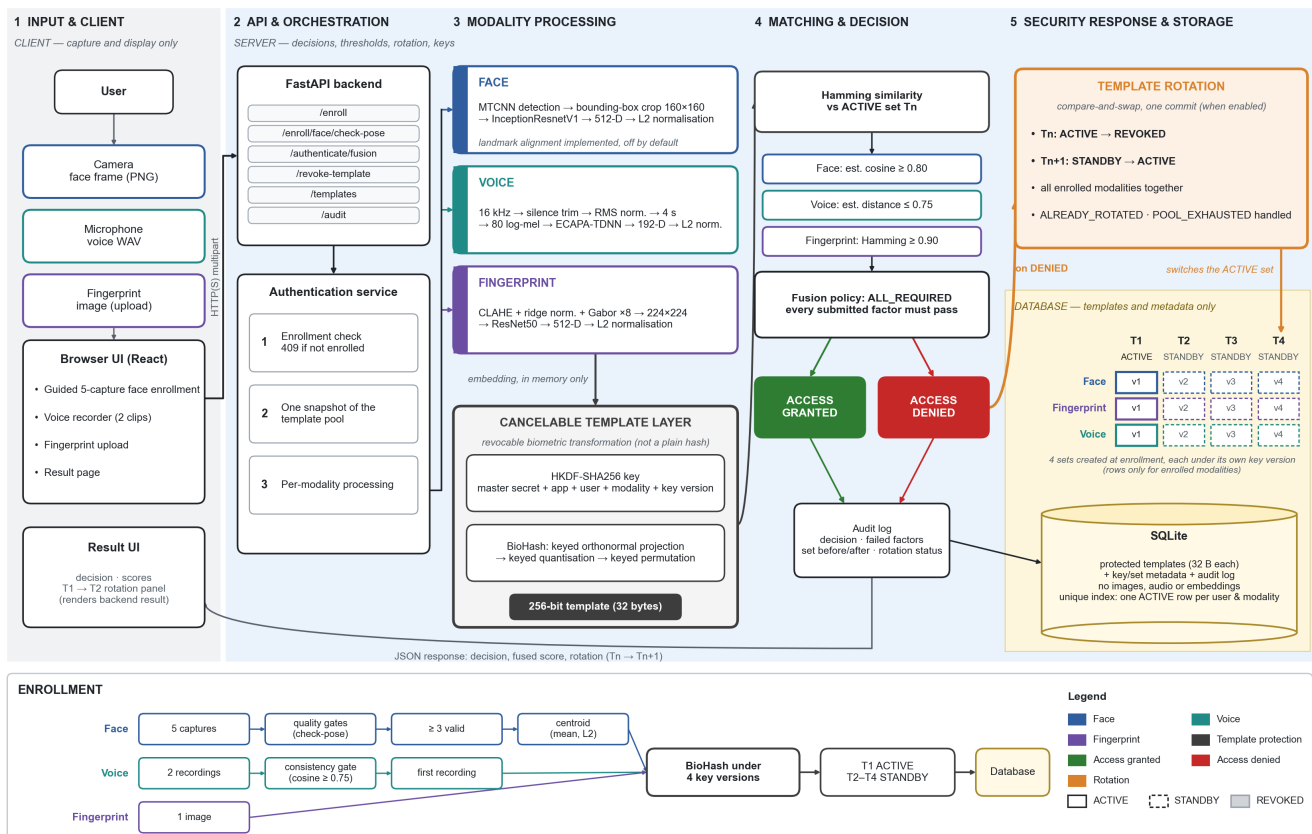


Figure: docs/system_architecture.pdf — the diagram to show on screen. Band 1 inputs + band 3 lanes + enrollment strip = Maanasvee; band 3 cancelable template layer + template pool = Reya; bands 2, 4, 5 (matching, decision, rotation, audit) = Sanya.

Corrections from the requested division (use these in the viva)

- **Face alignment is OFF in the running system.** The deployed face path crops MTCNN's bounding box to 160×160; the 5 landmarks are used only for enrollment quality checks. Landmark alignment is implemented and was evaluated but is not the default (setting FACE_ALIGNMENT=bbox).
- **Face "cosine" and voice "Euclidean distance" are calibrated estimates.** The backend never compares embeddings; it compares two 256-bit templates and converts the Hamming similarity into an estimated cosine or distance.
- **Face 0.80 and voice 0.75 are CONFIGURATION values requested by the teacher** (THRESHOLD_SOURCE = TEACHER_REQUESTED_BASELINE), not experimentally optimised thresholds.
- **OR is not a deployable fusion policy.** The code implements ALL_REQUIRED (default), WEIGHTED and AT_LEAST_TWO; OR appears only as a comparison rule in the evaluation.
- **Fingerprint uses Hamming similarity directly** (≥ 0.90), a default value, not a data-selected threshold.
- **Template rotation after failure is off by default** (TEMPLATE_ROTATION_ON_FAILED_AUTH); it is enabled for the demo.
- **Per-modality scores reach the result page only when DEBUG_SCORES=true.** In production mode the response carries only the decision and one fused score.
- **Voice templates come from the first of the two enrollment recordings;** the second is a consistency check.
- **Iris** exists in the code but has no trained model and is not part of the system you present.

Part 2 — Maanasvee Khetan: datasets, training, preprocessing & feature extraction

Maanasvee opens the technical story: which data we used, how the models were trained, and how a raw face, voice or fingerprint becomes the embedding that Reya's protection layer receives.

Topic 1 — Datasets

	Face	Voice	Fingerprint
Name	LFW (Labeled Faces in the Wild), funneled, via scikit-learn	VoxCeleb1 Indian-celebrity subset (Kaggle gaurav41/voxceleb1-audio-wav-files-for-india-celebrity)	SOCOFing (Kaggle ruizgara/socofing)
Identities / samples	fine-tuning: 62 people, 3,023 images (≥ 20 each); held-out evaluation: 1,618 other people, 6,141 images	24 speakers, 4,857 WAV clips	600 people $\times$ 10 fingers = 6,000 real prints + altered versions
Data type	web photos (RGB)	interview audio (WAV)	fingerprint images
Split	70 / 15 / 15 % per person, seed 42 (same people in every split)	3,389 / 717 / 751 clips; same 24 speakers in every split	420 / 90 / 90 people (subject-disjoint), seed 42
Preprocessing	MTCNN, bounding-box crop 160 $\times$ 160	16 kHz, trim, RMS, 4 s, 80 log-mel	CLAHE, ridge norm, Gabor $\times 8$, 224 $\times$ 224
Augmentation	none	none (options exist, disabled)	rotation, shift, scale, brightness, noise, blur, elastic
Use	fine-tuning + held-out evaluation	training + testing	fine-tuning + testing

Why these: free for non-commercial research, labelled by identity, standard in their fields (results are comparable), small enough for our hardware.

Evaluation protocol and dataset limits (REAL DATA)

- **Face:** held-out LFW identities (never used for fine-tuning); first image of each person = enrolled reference, the others = genuine probes; 50 random impostor probes per person, each templated under the target's key (seed 20260925): 4,523 genuine / 80,900 impostor comparisons.
- **Voice:** the 24-speaker test split, same enrollment-vs-probe protocol, all impostors: 727 genuine / 17,273 impostor.
- **Fingerprint:** subject-level protocol on 900 test images (all 404,550 pairs); different fingers of one person count as genuine, which makes it hard.
- **Limits:** LFW is web photos, not webcam captures; voice has only 24 speakers, shared across splits (closed-set); no demographic data; no real-user study yet (FUTURE WORK).

Topic 2 — Training

	Face	Voice	Fingerprint
Model	InceptionResnetV1 (facenet-pytorch)	ECAPA-TDNN	ResNet50 + projection head
Pretrained or trained?	Pretrained on VGGFace2, then fine-tuned : only block8, last_linear, last_bn + an ArcFace head were trained. It was not trained from scratch.	Trained from scratch in this project	ImageNet-pretrained backbone, fine-tuned on SOCOFing; conv1, bn1, layer1 frozen
Objective / loss	ArcFace (s 30, m 0.5) + cross-entropy	ArcFace (s 30, m 0.5) + cross-entropy	ArcFace (s 64, m 0.5, label smoothing 0.1) + hard-negative term from epoch 10
Optimizer, LR	Adam, 1e-4, no schedule	AdamW, 1e-3, weight decay 1e-4, cosine	AdamW, 3e-4, weight decay 1e-4, 3-epoch warm-up + cosine
Epochs / batch	10 / 32	up to 30, early stop patience 5 / 64	up to 30, patience 8, min 10 / 16 people × 4
Validation	15 % split (val accuracy)	15 % split (val loss)	15 % subjects (val EER)
Checkpoint	models/face/saved/face_embedder.pt	models/voice/saved/voice_embedder.pt	models/fingerprint/saved/fingerprint_embedder.pt
Result (REAL DATA)	held-out protected EER 8.50 %	test EER 2.29 %	EER 30.77 %, AUC 0.7644 (subject-level, 900 images)

Simple explanation: ArcFace is a training loss that pulls one person's embeddings together and pushes different people apart by angle. The classification head is discarded after training; only the embedding network is used. The notebooks did not save epoch-by-epoch logs for the deployed face and fingerprint models, so those are NOT AVAILABLE — don't invent them.

Topic 3 — Embeddings

What to say

An embedding is a fixed-length list of numbers that a network produces to describe an input. We use 512 numbers for face and fingerprint and 192 for voice, and we scale each to length 1. Two captures of the same person never give identical numbers — light, pose and noise change a little — but they land close together, while different people land far apart.

Simple explanation: the network summarises "who this is" as a point in space; same person → nearby points.

Key points to remember

- Face 512 · Voice 192 · Fingerprint 512, all L2-normalised (length 1).
- Raw pixels are a bad comparison: two photos of one person differ in thousands of pixels.
- Embeddings are much smaller: 512 numbers vs $160 \times 160 \times 3 = 76,800$ pixel values.

Likely panel questions — short answers

- **Why not compare raw images?** — pixels change with light and pose; embeddings keep identity.

Topic 4 — Face pipeline (current code)

```

Camera frame (PNG)
  ↓ MTCNN detector → box + confidence + 5 landmarks (enrollment: exactly one face, confidence ≥ 0.90;
                                                    login: the largest face)
  ↓ landmarks → quality checks only (roll, yaw) (alignment implemented but OFF)
  ↓ crop the bounding box, resize to 160 × 160 × 3
  ↓ pixel scaling (value - 127.5) / 128
  ↓ InceptionResnetV1 → 512 numbers → L2 normalisation
  ↓ to Reya's cancellable layer

```

- **Why each step:** detection finds the face so the background is ignored; the crop and fixed 160×160 size give the network the input shape it was trained on; pixel scaling matches the training normalisation; L2 normalisation makes only the direction of the embedding matter.

Topic 5 — Face alignment (status: implemented, not deployed)

- **Landmarks** are 5 points MTCNN returns: left eye, right eye, nose tip, left and right mouth corner.
- **Alignment** uses them to rotate, uniformly scale and shift the face so the eyes and mouth land on fixed pixel positions (a similarity transform). *Simple: straighten and centre every face the same way.*
- **Why it helps:** a tilted or off-centre face becomes a standard view, so the network sees more consistent input.
- **Current status:** implemented (FacePreprocessorAligned, mode FACE_ALIGNMENT=similarity) but the default is bbox. In the deployed mode the landmarks only feed the tilt/turn checks.
- **Measured (REAL DATA):** with a model retrained on aligned faces, protected EER fell 8.67 % → 7.18 %, mean eye-line tilt 2.89° → 0.98°, and 30° head-rotation EER 13.0 % → 6.75 %. Aligned input with the old model was worse (+1.46 points), so switching needs the retrained model and re-enrollment of every user.

Topic 6 — Five-capture enrollment and quality gates

Capture	What happens	Accepted if	If rejected
1 → 5 (each)	one camera frame → POST /enroll/face/check-pose → detection + gates; nothing stored	exactly one face, confidence ≥ 0.90, sharpness ≥ 25, face ≥ 15 % of frame height, centre offset ≤ 0.35, roll ≤ 20°, yaw ratio ≤ 0.20	a message (e.g. "too blurry"); retake; the counter does not move
After 5 VALID	"Enroll Face" unlocks; all 5 sent to POST /enroll in one request	backend re-checks: at least 3 must be VALID	fewer than 3 → nothing stored
Completion	embed the valid ones, average into one centroid, L2-normalise; per-pose embeddings deleted	—	—

Why the gates: each rejects a capture the model would embed badly — no face or several faces, a low-confidence detection, blur (Laplacian variance below 25), a face too small or off-centre, or a head tilted or turned too far. **Bug fix (relevant to the final behaviour):** the next slot is always the first empty one, and a click is ignored while a capture is still being checked — previously a double-click could skip a slot and leave the counter stuck at 4 of 5. **Why five:** averaging several captures gives a steadier "typical" face than one moment.

Topic 7 — Face embedding

What to say

InceptionResnetV1 takes the 160 by 160 crop, scaled to roughly -1 to +1, and outputs 512 numbers, which we normalise to length 1. At enrollment we average the valid captures into one embedding. At this point the raw face has become a numerical embedding — and that embedding goes straight into Reya's cancellable-template layer.

Simple explanation: the photo is now 512 numbers that describe the person, and that is all the next layer receives.

Key points to remember

- Model InceptionResnetV1, 512-D, expects 160×160×3 with $(x - 127.5)/128$, L2-normalised output.
- Never stored — the embedding goes to BioHash in memory.

Likely panel questions — short answers

- **What happens to the embedding next?** — BioHash turns it into four 256-bit templates; the embedding is deleted.

Topic 8 — Voice pipeline

WAV → resample 16,000 Hz, mono → trim near-silence (energy 2 %) → RMS loudness to 0.1
→ centre 4.0 s (zero-pad if short) → 80-band log-mel (25 ms window, 10 ms hop)
→ ECAPA-TDNN → 192 numbers → L2 normalisation → Reya's cancellable layer
Enrollment: 2 recordings; cosine of their embeddings ≥ 0.85 excellent, 0.75–0.84 good (enrolled),
0.60–0.74 fair (warning, user may continue), < 0.60 rejected; templates from the FIRST recording.

Simple explanation: a log-mel spectrogram is a picture of the voice's energy at each pitch band over time, spaced like human hearing. Resampling and mono make every recording comparable; trimming silence and fixing loudness stop quiet pauses or volume from changing the embedding; a fixed 4 s clip gives the network a fixed input size.

Topic 9 — Fingerprint pipeline

Uploaded image (no physical scanner) → grayscale → CLAHE (clip 3.0, 8×8 tiles) → ridge normalisation
→ 3×3 Gaussian blur → Gabor filter bank, 8 orientations → min-max to 0–255 → 224 × 224
→ ImageNet normalisation → ResNet50 + head → 512 numbers → L2 normalisation → Reya's cancellable layer
Enrollment: a single image.

Simple explanation: CLAHE boosts contrast locally; Gabor filters respond to stripes at one angle, so eight of them make the ridge pattern stand out in every direction. The blur removes sensor noise first; 224×224 with ImageNet normalisation is the input the ResNet50 backbone expects.

Maanasvee → Reya handoff

Say: "So every face, voice and fingerprint ends up as an embedding — 512 or 192 numbers — and that embedding is never saved. But an embedding is permanent: if it leaked, it would identify you forever. Reya will now explain how we turn it into a protected template that can be revoked and replaced."

Part 3 — Reya Oberoi: cancelable biometric & template architecture

Reya explains what happens to the embedding Maanasvee produced: how it becomes a keyed, revocable 256-bit template, how templates are organised in four sets, and what is (and is not) stored. Sanya's matching then uses these templates.

Topic 1 — Why cancellable biometrics?

What to say

A password can be changed after a data breach, but your face, voice and fingerprint cannot. A traditional system stores a template of your biometric — typically the embedding itself. If that database leaks, the attacker keeps a permanent copy, and the same template can link you across every service that uses it. Cancellable biometrics means we never store that representation. We store a version transformed with a secret key. If it leaks, we switch to another key and issue a new, unrelated template made from the same biometric.

Simple explanation: the stored thing becomes replaceable like a password, even though your body is not.

Key points to remember

- Password: can be changed. Biometric: cannot be changed.
- Traditional template: if compromised, it cannot be revoked; it is also linkable across systems.
- Cancellable template: a new transformed representation can be issued from the same biometric.
- We protect the stored biometric representation, not the network channel (that is HTTP(S)).

Likely panel questions — short answers

- **What exactly are you protecting?** — the biometric representation stored in the database.
- **Isn't encryption enough?** — an encrypted template must be decrypted to be compared, so the real template exists at match time; ours is compared in its protected form.

Topic 2 — From embedding to cancellable template

What to say

Maanasvee's layer hands me an embedding. First it is scaled to length 1. Then the server derives a secret key for this application, user, modality and key version. BioHash uses that key to turn the embedding into 256 bits — the cancellable template — and that is what we store.

Simple explanation: numbers describing you go in; a scrambled, replaceable 256-bit code comes out.

Key points to remember

- Embedding (in): 512 numbers for face and fingerprint, 192 for voice.
- L2 normalisation: divides by the vector's length so only direction matters.
- Key: HKDF-SHA256 from MASTER_SECRET + application + user + modality + key version.
- Out: 256 bits = 32 bytes, stored in one template set.

Likely panel questions — short answers

- **Why normalise first?** — so the same person's samples differ only in direction, which the projection preserves.

Embedding (from Maanasvee)	in: 512 or 192 real numbers
↓ L2 normalisation	out: same numbers, length 1 (only direction matters)
↓ key derivation	in: master secret + app + user + modality + key version
	out: 3 seeds (projection, threshold, permutation)
↓ BioHash	in: unit embedding + seeds
	out: 256 bits
↓ storage	32 bytes in one set (T1...T4); the embedding is then deleted

Topic 3 — HKDF-SHA256 key derivation and binding

HKDF (HMAC-based Key Derivation Function) turns one secret into many independent keys. *Simple: one master secret plus a label deterministically produces a separate secret recipe for every user, factor and version.* Our code (template_protection/hkdf_keys.py::derive_key):

```
context = "application | user | modality | key_version"
salt = SHA-256(context)
projection_seed = HKDF-SHA256(MASTER_SECRET, salt, info = "template_protection/projection")
threshold_seed = HKDF-SHA256(MASTER_SECRET, salt, info = "template_protection/threshold")
permutation_seed = HKDF-SHA256(MASTER_SECRET, salt, info = "template_protection/permutation")
```

- **Binding:** because application, user, modality and key version are all in the context, a template is tied to exactly one of each; changing any one gives unrelated seeds.
- **MASTER_SECRET** lives only in the server environment (.env), never in the database; keys are re-derived when needed, never stored.
- Revoked key versions are never reused (the next version continues after the highest ever used).

Topic 4 — BioHash, exactly as implemented

Our function `generate_template` (template_protection/biohash.py) does four steps:

- **L2 normalise** the embedding: $\hat{x} = x / \|x\|$.
- **Keyed orthonormal projection:** $y = W \cdot \hat{x}$, where W has 256 orthonormal rows built by a QR decomposition of a random Gaussian matrix seeded by the projection seed. For voice (192 numbers) the rows are built in two orthonormal blocks. *Simple: look at the embedding from 256 secret, independent directions.*
- **Keyed quantisation:** $b_i = 1$ if $y_i > t_i$, else 0, with thresholds t_i drawn from a normal distribution of spread $0.5 \times$ the spread of y (seeded by the threshold seed). *Simple: turn each number into a yes/no bit.*
- **Keyed permutation:** $\text{template} = \pi(b)$, a shuffle of the 256 bit positions from the permutation seed. *Simple: scramble the order.*

Comparison (used by Sanya): Hamming similarity $H(a, b) = 1 - (1/256) \cdot \sum [a_i \neq b_i]$ — the fraction of equal bits.

Question	Answer
Is it SHA-256?	No. SHA-256 is used only inside key derivation (HKDF-SHA256, and a SHA-256 hash of the context as HKDF's salt). The template comes from projection + thresholds + permutation.
Is it ordinary cryptographic hashing?	No. A cryptographic hash changes completely if the input changes slightly, so two photos of the same person would never match. BioHash keeps similar inputs similar.
What is the input?	The L2-normalised embedding and the key (three seeds).
What role does the key play?	It fixes the secret projection, thresholds and shuffle. Without the master secret you cannot make a comparable template.
What role does the version play?	It is part of the key context, so each version gives different seeds and therefore an unrelated template from the same embedding.
What makes it cancellable?	A new version gives a new template; old versions can be retired and are never reused.
Evidence that versions differ (REAL DATA)	The same face embedding under different set keys agrees on 49.9 % of bits (0.499 ± 0.032) — what two random bit strings give.

Topic 5 — The 256-bit template

What to say

The template is 256 bits, which is 32 bytes. It is stored as packed bytes in the `protected_template` column. It is derived from the embedding but it is not the embedding: 256 yes/no answers instead of 512 real numbers, and it contains no image, audio or embedding values.

Simple explanation: a 32-byte code that can be compared, but that is useless without the server's secret and can be replaced.

Key points to remember

- 256 bits = 32 bytes per template; the column `output_bits` = 256 (CONFIGURATION).
- Why 256: a measured experiment (documented in `backend/config.py`) showed 128 bits lost about 7 percentage points of EER for face and voice; 256 recovered most of it.
- $256 \leq 512$ (face, fingerprint) → one orthonormal projection block; voice (192) needs two blocks.

Likely panel questions — short answers

- **Why 256 bits?** — we measured it: 128 bits discarded identity information, 256 kept most of it.
- **Does it contain the biometric?** — no image, audio or embedding values — only keyed, thresholded bits.

Topic 6 — The four template sets

At enrollment the backend makes **four** templates per enrolled modality from the same embedding, each under a different key version. They are generated during enrollment, not later.

T1	ACTIVE	└─ Face (key v1)	└─ Fingerprint (key v1)	└─ Voice (key v1)
T2	STANDBY	└─ Face (key v2)	└─ Fingerprint (key v2)	└─ Voice (key v2)
T3	STANDBY	└─ Face (key v3)	└─ Fingerprint (key v3)	└─ Voice (key v3)
T4	STANDBY	└─ Face (key v4)	└─ Fingerprint (key v4)	└─ Voice (key v4)

(only enrolled modalities have rows; a face-only user has Face rows only)

Status	Technical meaning	Simple meaning
ACTIVE	the one set compared during authentication (is_active = true)	the key currently in the lock
STANDBY	prepared at enrollment, never compared, waiting to replace the active set	spare keys in the drawer
REVOKED	retired permanently; its key version is never reused	a key that has been destroyed

- Pool size 4 is the setting TEMPLATE_POOL_SIZE; so 3 replacements are possible before re-enrollment.
- A modality enrolled later joins the existing live sets, so every set holds every enrolled modality.

Topic 7 — Database storage (verified)

Stored	NOT stored
32-byte protected templates (protected_template)	raw face images, voice recordings, fingerprint images
key_version, template_set_version, status, timestamps, revoked_reason	embeddings (the face centroid is deleted right after the four templates are generated)
audit metadata: decision, factors, scores, set before/after, rotation status	landmarks
an HKDF fingerprint of the master secret (detects a changed secret)	the master secret, any key or seed (keys are re-derived when needed)

Tables: users, protected_templates (one row per user × application × modality × set), audit_logs, enrollment_events, master_secret_fingerprints. A partial unique index allows at most one ACTIVE row per user, modality and application.

Why no images or embeddings: they are permanent and linkable; if the database leaks, only revocable templates are exposed.

Topic 8 — Revocation and replacement

T1 ACTIVE
↓ T1 compromised (manual authorised revoke) – or a failed login when rotation is on (Sanya's section)
T1 REVOKED
↓ the oldest STANDBY set is promoted
T2 ACTIVE
↓ the real user logs in: live sample → template under key v2 → compared with T2 → matches

- **No raw biometric is needed:** T2 already exists from enrollment; revocation only changes statuses.
- **Different from re-enrolling a new person:** T2 was made from the same user's own enrollment embedding; nothing about the user's identity changes, only the key.
- **The legitimate user still works (REAL DATA):** genuine similarity was unchanged after 3 rotations (face 0.761 → 0.766), while a revoked template vs the new active one agrees on 0.500 of bits with 0 acceptances.
- **Manual revoke (/revoke-template)** needs a successful biometric check of every active factor first.
- **Failed samples are never stored:** rotation creates no template; a test checks the promoted set is byte-identical to the one made at enrollment.

Topic 9 — Unlinkability, revocability and security properties

Property	What it means	Evidence in our project
Revocability	a compromised template can be retired and replaced	REVOKED sets are never compared; revoked vs new active $\approx$ 0.500 bit agreement, 0 acceptances (REAL DATA)
Unlinkability	two templates of the same person under different keys can't be matched to each other	cross-key agreement 0.499 ± 0.032 ; $D \leftrightarrow \text{sys}$ (bin 1/64): face 0.018 (small but above the null floor 0.013), voice 0.020 and fingerprint 0.014 (not distinguishable from the null) — REAL DATA
Performance kept	protection costs little accuracy	face EER 7.12 % raw $\rightarrow$ 8.50 % protected; voice 2.21 % $\rightarrow$ 2.32 % (REAL DATA)
No raw storage	only templates and metadata are stored	database schema (CONFIGURATION)
Irreversibility	a template can't be turned back into the biometric	not formally claimed: lossy and keyed, but no proof or attack study (FUTURE WORK)
Trust assumption	the master secret must stay secret	lives only in the server environment; fingerprint checked at startup

Reya $\rightarrow$ Sanya handoff

Say: "So every biometric is stored only as protected 256-bit templates, in four sets with different keys, and only one set is active at a time. Nothing we store is an image, a recording or an embedding. Sanya will now show how the system uses that active set when someone tries to log in — how the templates are compared, how the final decision is made, and what the system does when a login fails."

Part 4 — Sanya Malik: authentication, matching, fusion & security response

Sanya closes the technical story: how a login compares a fresh template with Reya's active set, how the per-factor decisions are fused into one answer, how the system responds to a failure, and what the results show.

Topic 1 — End-to-end authentication

```
User picks enrolled factors, captures them (frontend: AuthenticatePage)
↓ POST /authenticate/fusion
Backend: every presented factor enrolled? no → 409 ENROLLMENT_REQUIRED (nothing evaluated)
↓ ONE read of the user's template sets (a snapshot for this attempt)
Per factor: preprocess → embedding → template under the ACTIVE row's key
↓ compare with the ACTIVE template → Hamming similarity → estimated metric
Per factor decision (threshold) → fusion policy ALL_REQUIRED → GRANTED / DENIED
↓ denied + rotation enabled → Tn → Tn+1
Audit row → JSON response → result page
```

- **The backend is the only decision point** (backend/services/authentication.py::authenticate_samples). Thresholds, fusion and rotation all run on the server.
- **The frontend** captures samples and displays the response; it never decides access or rotation.
- **Matching never uses embeddings directly:** the live sample is templated with the same key version as the stored ACTIVE template, then the two 256-bit templates are compared.

Topic 2 — Hamming similarity and the estimated metrics

- **Hamming similarity** = fraction of the 256 bits that are equal (1.0 identical, ≈ 0.5 unrelated).
- BioHash keeps similar embeddings similar, so the Hamming similarity tracks how close the embeddings were. A **calibration curve** (template_protection/metric_estimation.py) converts it into an **estimated cosine**.
- The curve was fitted on synthetic vector pairs through the real transform — **SYNTHETIC calibration** ($R^2 \approx 0.9999$). It was then checked on real embeddings — **REAL DATA**: face estimate error ≈ 0.07 on genuine pairs. The synthetic fit is not a biometric-performance result.
- Voice converts the estimated cosine into a distance: $d = \sqrt{2 - 2 \cdot \hat{c}}$ (exact for length-1 vectors).

Topic 3 — Face decision rule

What to say

The face image is detected with MTCNN, the largest face is cropped to 160 by 160 pixels, and InceptionResnetV1 turns it into 512 numbers. That embedding is templated with the active face key and compared with the stored face template. The comparison gives an estimated cosine similarity; 0.80 or higher passes.

Simple explanation: cosine similarity says how much two face descriptions point the same way; higher means more similar.

Key points to remember

- estimated cosine $\geq 0.80 \rightarrow$ pass; $< 0.80 \rightarrow$ fail (DERIVED: $\approx$ at least 210 of 256 bits equal).
- Cosine suits face because the model was trained with ArcFace (angle-based) and embeddings have length 1.
- No quality gates at login (they run only at enrollment). Authentication takes the largest detected face.
- **0.80 is a teacher-requested CONFIGURATION value** (THRESHOLD_SOURCE = TEACHER_REQUESTED_BASELINE), not an experimentally optimised threshold.
- Measured on held-out web photos (REAL DATA): false rejects 81.5 %, false accepts 0.0025 % at 0.80 — very strict.

Likely panel questions — short answers

- **Is it a real cosine?** — a calibrated estimate from the Hamming comparison — embeddings are never stored.

Topic 4 — Voice decision rule

What to say

The recording is resampled to 16 kHz, trimmed and normalised, 4 seconds are turned into an 80-band log-mel spectrogram, and ECAPA-TDNN produces 192 numbers. It is templated with the active voice key and compared. The result is expressed as an estimated Euclidean distance; 0.75 or less passes.

Simple explanation: Euclidean distance is how far apart two voice descriptions are; smaller means more alike, so lower is better.

Key points to remember

- estimated distance $\leq 0.75 \rightarrow$ pass; $> 0.75 \rightarrow$ fail.
- DERIVED: for length-1 vectors $d = \sqrt{(2 - 2 \cdot \cos)}$, so 0.75 equals estimated cosine ≥ 0.719 ($\approx$ at least 201 of 256 bits).
- **0.75 is a teacher-requested CONFIGURATION value**, not experimentally optimised.
- Noise, a different microphone or a quiet short clip move the embedding, so the distance grows and a genuine speaker can fail — an ordinary failure, not a bug and not proof of an attack.
- Measured (REAL DATA): voice EER 2.32 %; at 0.75, false rejects 23.8 %, false accepts 0.26 %.

Likely panel questions — short answers

- **Why is lower better for voice?** — it is a distance: 0 = identical, larger = more different.

Topic 5 — Fingerprint decision rule

What to say

The fingerprint image is enhanced, resized to 224 by 224 and turned into 512 numbers by ResNet50. It is templated with the active fingerprint key. Unlike face and voice, fingerprint uses the Hamming similarity directly: at least 90 percent of the bits must match.

Simple explanation: count how many of the 256 bits agree; 90 % or more passes.

Key points to remember

- Hamming similarity $\geq 0.90 \rightarrow$ pass (DERIVED: $\approx$ at least 231 of 256 bits).
- 0.90 is the backend's default MATCH_THRESHOLD (CONFIGURATION); no fingerprint calibration file exists.
- Fingerprint is optional in the deployed flow.

Likely panel questions — short answers

- **Why no estimated metric?** — no calibrated fingerprint threshold exists, so the raw Hamming rule is used.

Topic 6 — Fusion, policies and fusion-scale normalisation

Fusion combines the per-factor results into one decision (fusion/policy.py::evaluate_fusion_policy). *Simple: one final yes/no from several checks.*

- **Fusion-scale normalisation:** every factor's score is put on one higher-is-better scale, the estimated cosine: face = $\hat{c}$ (threshold 0.80); voice = $1 - d^2/2$ (threshold 0.7188); fingerprint = its calibration curve (threshold 0.9384). A distance is never averaged with a similarity.
- **Score-level fusion:** the fused score is the equal-weight average of the presented factors' normalised scores.

Policy	Rule	Status in code
ALL_REQUIRED	every presented factor must pass its own threshold	default
WEIGHTED	fused score $\geq$ mean threshold, and no presented score below the floor (0.0)	implemented
AT_LEAST_TWO	exactly three factors presented; at least two pass	implemented
OR	any one factor passes	evaluation-only comparison rule; not a deployable policy

Example	What the code does	Result
A — all presented factors pass	every factor $\geq$ its threshold	ACCESS GRANTED
B — face fails, voice and fingerprint pass	ALL_REQUIRED needs every presented factor	ACCESS DENIED (rotation if enabled)
C — only face presented	only presented factors are evaluated; face alone decides	face pass $\rightarrow$ GRANTED; fail $\rightarrow$ DENIED
D — a presented factor is not enrolled	checked before anything is evaluated	409 ENROLLMENT_REQUIRED; no score, no rotation

Topic 7 — Access granted / denied (security response)

- **Granted:** templates unchanged, no rotation; one audit row (rotation status NOT_APPLICABLE); the response carries the decision, the fused similarity, the active set version and the display name. Per-modality scores are included only when DEBUG_SCORES=true (demo mode).
- **Denied:** no display name is returned; the audit row records which factors failed; if rotation is enabled the next step runs.

```
Verification → a presented factor fails → ACCESS DENIED
↓ backend: rotation enabled? no → nothing changes
↓ yes → compare-and-swap rotation (ROTATED / ALREADY_ROTATED / POOL_EXHAUSTED / ROTATION_FAILED)
↓ audit row (failed factors, set before/after, rotation status)
↓ result page: decision + "Authentication failed – template rotation triggered" + T1 → T2 table
```

Topic 8 — Template rotation after a failed login

```
T1 ACTIVE → failed authentication → T1 REVOKED, T2 ACTIVE    (face, voice, fingerprint together)
next failure      → T2 REVOKED, T3 ACTIVE
next failure      → T3 REVOKED, T4 ACTIVE
next failure      → POOL_EXHAUSTED (nothing changes)
```

- **Why:** a failed attempt may mean the active template is being attacked; moving to a fresh key makes any copy of the old template useless (Reya's revocation mechanism).
- **Who triggers it:** the backend (rotate_after_failed_authentication → crud.rotate_active_set_if_current), only from its own ACCESS_DENIED decision and only if TEMPLATE_ROTATION_ON_FAILED_AUTH is on.
- **Attacker sample never stored:** rotation creates no template; the promoted set was made at enrollment.
- **All modalities move together:** the whole set is retired and the next whole set promoted in one commit.
- **Wording:** the backend only knows verification failed — say "authentication failed", never "attack detected".

Topic 9 — Concurrency

```
Without protection:   Request A: T1 → T2      Request B: T1 → T3    (a set is wasted)

Current fix (compare-and-swap):
  UPDATE ... SET REVOKED WHERE set = T1 AND status = 'ACTIVE' ← takes the database write lock first
  Request A: finds T1 ACTIVE → revokes T1, promotes T2, checks row counts, commits once
  Request B: re-checks → 0 rows (T1 already revoked) → changes nothing → ALREADY_ROTATED
```

Simple explanation: "change it only if nobody changed it first". In tests with 8 or 16 simultaneous failures the result is always exactly one rotation (SOFTWARE TEST). Each attempt also reads the template pool once, so it never sees a half-rotated state. After T4, POOL_EXHAUSTED: nothing changes and the user still logs in on T4.

Topic 10 — Audit logging

Every attempt writes one audit_logs row: attempt ID, time, user, building, presented / enrolled / verified / failed factors, per-factor scores and thresholds, fused score, policy, decision, set version, key versions, active_set_before, active_set_after, template_rotation_triggered, template_rotation_status, latency. It proves what was decided and whether a rotation happened. **It never stores** images, audio, embeddings, templates or keys.

Topic 11 — Authentication results and limitations

Measure	Value	Label
Face protected EER (held-out LFW, 4,523 / 80,900)	8.50 % (raw 7.12 %), ROC-AUC 0.9641	REAL DATA
Voice protected EER (727 / 17,273)	2.32 % (raw 2.21 %), ROC-AUC 0.9967	REAL DATA
Face at 0.80	FRR 81.5 %, FAR 0.0025 %	REAL DATA
Voice at 0.75	FRR 23.8 %, FAR 0.26 %	REAL DATA
Face + voice ALL_REQUIRED (chimeric users)	FRR 84.9 %, FAR 0 %	REAL DATA
Face + voice WEIGHTED (chimeric)	FRR 59.2 %, FAR 0 %	REAL DATA
Face + voice OR (evaluation rule only)	FRR 19.4 %, FAR 0.14 %	REAL DATA
Score-level mean(face, voice)	EER 1.67 %, AUC 0.9989	REAL DATA
Three factors: AT_LEAST_TWO	FRR 22.8 %, FAR 0.024 %	REAL DATA
Face, FAR $\leq$ 1 % point chosen on a development split	test FAR 0.83 %, FRR 22.9 % — not adopted	REAL DATA
Rotation under 8 / 16 simultaneous failures	exactly 1 rotation	SOFTWARE TEST
Backend automated tests	657 passing — software tests, not authentication trials	SOFTWARE TEST

- **Chimeric users:** fusion results pair LFW faces with VoxCeleb voices as simulated people; this assumes the factors are independent.
- **Thresholds are not tuned:** 0.80 and 0.75 are configuration values; choosing new ones needs an agreed FAR target and real users (FUTURE WORK).
- **No real-user study yet** (FUTURE WORK); LFW is web photos; the voice test set has 24 speakers.

Sanya's closing system summary

Say: "So, to close the loop: Maanasvee showed how a live face, voice or fingerprint becomes an embedding. Reya showed how that embedding becomes four protected 256-bit templates under different keys, with only one set active. At login, our server templates the new sample with the active key, compares bits, converts that into an estimated score, applies the configured thresholds and the ALL_REQUIRED policy, and grants or denies access — and after a denial it can safely rotate every factor to the next template set without storing anything from the attempt. Raw biometric, embedding, protected template, decision, rotation: that is the whole system."

Part 5 — Who speaks about what

PRIMARY = presents it. MUST UNDERSTAND = can answer a question on it. — = not their section. Every topic has exactly one PRIMARY speaker.

Topic	Maanasvee	Reya	Sanya
Datasets, sizes, splits, evaluation protocol	PRIMARY	—	MUST UNDERSTAND
Training, pretrained vs from scratch	PRIMARY	—	—
Face detection, landmarks, alignment status	PRIMARY	—	MUST UNDERSTAND
Face quality gates, five-capture enrollment	PRIMARY	—	—
Voice and fingerprint preprocessing	PRIMARY	—	MUST UNDERSTAND
Embeddings and dimensions	PRIMARY	MUST UNDERSTAND	MUST UNDERSTAND
Why cancellable biometrics	—	PRIMARY	MUST UNDERSTAND
HKDF keys, binding, versions	MUST UNDERSTAND	PRIMARY	—
BioHash (projection, quantisation, permutation)	MUST UNDERSTAND	PRIMARY	MUST UNDERSTAND
256-bit template, storage	MUST UNDERSTAND	PRIMARY	MUST UNDERSTAND
Four template sets, statuses, revocation	—	PRIMARY	MUST UNDERSTAND
Unlinkability, security properties	—	PRIMARY	—
Authentication flow, backend authority	—	MUST UNDERSTAND	PRIMARY
Hamming similarity, estimated metrics	—	MUST UNDERSTAND	PRIMARY
Thresholds (0.80 / 0.75 / 0.90)	MUST UNDERSTAND	MUST UNDERSTAND	PRIMARY
Fusion policies, normalisation	—	—	PRIMARY
Rotation, concurrency, POOL_EXHAUSTED	—	MUST UNDERSTAND	PRIMARY
Audit, authentication results, limitations	—	—	PRIMARY

Part 6 — Important numbers (verified)

Item	Value	Label	Who must know
Face threshold	estimated cosine ≥ 0.80 ($\approx 210/256$ bits)	CONFIGURATION (teacher-requested)	Sanya (all)
Voice threshold	estimated distance ≤ 0.75 (= est. cosine ≥ 0.719 , $\approx 201/256$ bits)	CONFIGURATION (teacher-requested)	Sanya (all)
Fingerprint threshold	Hamming ≥ 0.90 ($\approx 231/256$), default	CONFIGURATION	Sanya
Fusion-scale thresholds	face 0.80 · voice 0.7188 · fingerprint 0.9384	DERIVED	Sanya
Default fusion	ALL_REQUIRED over presented factors	CONFIGURATION	Sanya
Embedding dimensions	face 512 · voice 192 · fingerprint 512	CONFIGURATION	Maanasvee (all)
Template	256 bits = 32 bytes	CONFIGURATION	Reya (all)
Template sets	4 per user (1 ACTIVE + 3 STANDBY) → 3 rotations	CONFIGURATION	Reya, Sanya
Face enrollment	5 captures sent, ≥ 3 must be valid → centroid	CONFIGURATION	Maanasvee
Face gates	confidence ≥ 0.90 , sharpness ≥ 25 , size ≥ 0.15 , centre ≤ 0.35 , roll $\leq 20^\circ$, yaw ≤ 0.20	CONFIGURATION	Maanasvee
Face input	$160 \times 160 \times 3$, $(x - 127.5)/128$	CONFIGURATION	Maanasvee
Voice enrollment	2 recordings; ≥ 0.75 enrolled, 0.60–0.74 warning, < 0.60 rejected	CONFIGURATION	Maanasvee
Audio	16 kHz mono, 4.0 s, 80 mel bands, 25 ms / 10 ms	CONFIGURATION	Maanasvee
Fingerprint input	224×224 , 8 Gabor orientations	CONFIGURATION	Maanasvee
Face training	62 identities, 3,023 images, Adam 1e-4, batch 32, 10 epochs	CONFIGURATION	Maanasvee
Voice training	24 speakers, 4,857 clips, from scratch, AdamW 1e-3, batch 64, ≤ 30 epochs	CONFIGURATION	Maanasvee
Fingerprint training	600 subjects, 420/90/90, AdamW 3e-4, ≤ 30 epochs	CONFIGURATION	Maanasvee
BioHash quantisation	thresholds with spread $0.5 \times$ data spread	CONFIGURATION	Reya
Key derivation	HKDF-SHA256, 3 seeds (projection, threshold, permutation)	CONFIGURATION	Reya
Cross-key agreement	0.499 ± 0.032 of bits ($\approx$ random)	REAL DATA	Reya
128 vs 256 bits	128 lost ≈ 7 EER points for face and voice	REAL DATA	Reya
Calibration curve fit	$R^2 \approx 0.9999$	SYNTHETIC	Sanya, Reya
Calibration on real embeddings	face genuine-pair error ≈ 0.07	REAL DATA	Sanya
Face results	EER 8.50 % protected (7.12 % raw); at 0.80 FRR 81.5 %, FAR 0.0025 %	REAL DATA	Sanya, Maanasvee
Voice results	EER 2.32 % protected (2.21 % raw); at 0.75 FRR 23.8 %, FAR 0.26 %	REAL DATA	Sanya, Maanasvee
Face + voice ALL_REQUIRED	84.9 % of genuine simulated attempts rejected	REAL DATA	Sanya
Alignment experiment	EER 8.67 % → 7.18 % (retrained model; not deployed)	REAL DATA	Maanasvee
Fingerprint model	EER 30.77 %, AUC 0.7644	REAL DATA	Maanasvee
Concurrency test	8 / 16 simultaneous failures → exactly 1 rotation	SOFTWARE TEST	Sanya
Automated tests	657 passing — software tests, not biometric trials	SOFTWARE TEST	all

Part 7 — Master system flow with speaker ownership

```
[MAANASVEE]    User input: camera frame / voice WAV / fingerprint image
↓
[MAANASVEE]    Preprocessing: face MTCNN + bbox crop 160x160 | voice 16 kHz, trim, RMS, 4 s, log-mel
                | fingerprint CLAHE + Gabor, 224x224
↓
[MAANASVEE]    Embedding: InceptionResnetV1 512 | ECAPA-TDNN 192 | ResNet50 512 → L2 normalised
↓
[REYA]         Key: HKDF-SHA256(master secret; app | user | modality | key version)
↓
[REYA]         BioHash: projection → quantisation → permutation → 256-bit template
↓
[REYA]         Enrollment: 4 sets (T1 ACTIVE, T2-T4 STANDBY), stored as 32-byte templates
↓
[SANYA]        Login: enrollment check → one snapshot of the pool → ACTIVE template per factor
↓
[SANYA]        Matching: Hamming similarity → estimated cosine / distance
↓
[SANYA]        Decisions: face ≥ 0.80 · voice ≤ 0.75 · fingerprint ≥ 0.90 (configuration values)
↓
[SANYA]        Fusion: ALL_REQUIRED → ACCESS GRANTED / ACCESS DENIED
                ↓ (denied, rotation enabled)
[SANYA]        Compare-and-swap rotation → ROTATED / ALREADY_ROTATED / POOL_EXHAUSTED
↓
[REYA+SANYA]   New ACTIVE set Tn+1 (all enrolled modalities)
↓
[SANYA]        Audit row → result page
```

Part 8 — Panel questions for Maanasvee

#	Question	Short answer	Detailed answer
1	What datasets did you use?	LFW, VoxCeleb1 subset, SOCOFing.	Face 62/3,023 (fine-tune) + 1,618/6,141 (evaluation); voice 24/4,857; fingerprint 600/6,000 real.
2	Why these datasets?	Free, labelled, standard.	Comparable with literature and small enough for our hardware.
3	How were they split?	70/15/15.	Face per person; voice by clip (same speakers); fingerprint by subject (420/90/90).
4	How was the face model evaluated?	On 1,618 unseen people.	Enrollment-vs-probe: 4,523 genuine / 80,900 impostor comparisons, 50 impostors per person.
5	What are the dataset limitations?	Web photos, 24 speakers.	LFW isn't webcam data; voice is closed-set; no demographics; no real-user study yet.
6	Did you train the face model?	Fine-tuned, not from scratch.	VGGFace2-pretrained; only block8, last_linear, last_bn + ArcFace head trained.
7	Did you train the voice model?	Yes, from scratch.	ECAPA-TDNN, AdamW 1e-3, batch 64, up to 30 epochs.
8	And fingerprint?	Fine-tuned from ImageNet.	ResNet50, early layers frozen, subject-disjoint split.
9	What loss did you use?	ArcFace.	An angular-margin loss that clusters each identity by angle.
10	What is an embedding?	Numbers describing identity.	A fixed-length vector (512 or 192) where the same person lands nearby.
11	Why not compare raw images?	Pixels vary with light and pose.	Embeddings keep identity and are much smaller.
12	Why do two captures differ slightly?	Light, pose, noise.	Still close in embedding space because the network learned identity.
13	What detects the face?	MTCNN.	Box, confidence and 5 landmarks.
14	What are landmarks?	5 key points.	Eyes, nose tip, mouth corners.
15	Does the running system align faces?	No — bounding-box crop.	Alignment implemented and evaluated, but off; landmarks only feed quality checks.
16	Why would alignment help?	Standardises pose.	Retrained aligned model: EER 8.67 % → 7.18 %.
17	Why not switch it on?	Needs a new model + re-enrollment.	Aligned input with the old model was worse.
18	Why five captures?	A steadier average.	At least 3 valid captures are averaged into one centroid.
19	When is a capture rejected?	When a gate fails.	No/multiple faces, confidence < 0.90, blurry, too small, off-centre, tilted or turned.
20	Why are the quality gates needed?	Bad captures make bad embeddings.	Each gate removes a known failure: blur, small or off-centre faces, head pose.
21	What input does the face model expect?	160×160×3.	Pixels scaled $(x - 127.5)/128$.
22	How is voice processed?	16 kHz → log-mel → ECAPA-TDNN.	Trim, RMS, 4 s, 80 mel bands, 192-D output.
23	Why two voice recordings?	Consistency check.	Embeddings must agree (≥ 0.75); templates from the first.
24	How is a fingerprint processed?	Ridge enhancement → ResNet50.	CLAHE, ridge norm, Gabor ×8, 224×224, 512-D.
25	Where do embeddings go?	Into BioHash, in memory.	Never stored; turned into templates (Reya) and deleted.

Part 9 — Panel questions for Reya

#	Question	Short answer	Detailed answer
1	What exactly are you protecting?	The stored biometric representations.	We store 256-bit BioHash templates made with secret, versioned keys instead of embeddings, so a leak can be revoked.
2	How does BioHash work?	Normalise, project, threshold, shuffle.	$\hat{x} = x/\ x\ $; $y = W \cdot \hat{x}$ (keyed orthonormal rows); $b = [y > t]$ (keyed thresholds); template = $\pi(b)$ (keyed permutation).
3	Is this SHA-256?	No.	SHA-256 is only inside HKDF key derivation. A hash would make two photos of the same person unmatchable.
4	Is it ordinary cryptographic hashing?	No.	Cryptographic hashes destroy similarity; BioHash keeps similar embeddings similar so matching works.
5	What is HKDF?	A standard key-derivation function.	HMAC-based; from one secret plus a context it produces independent keys — here three seeds.
6	What goes into the key?	Secret + app + user + modality + version.	Salt = SHA-256 of that context; three HKDF outputs with different info labels.
7	Why an orthonormal (QR) projection?	Independent, non-redundant directions.	Orthonormal rows keep angles between embeddings, so similar faces stay similar; voice uses two blocks.
8	How are the thresholds generated?	From the threshold seed.	Normal distribution with spread $0.5 \times$ the projected values' spread.
9	Why permute the bits?	To hide which direction each bit came from.	A keyed shuffle of the 256 positions.
10	Why 256 bits?	Measured trade-off.	128 bits lost ≈ 7 EER points for face and voice; 256 recovered most of it.
11	How many bytes is a template?	32.	256 bits packed into 32 bytes in protected_template.
12	Does the template contain the biometric?	No image, audio or embedding.	Only 256 keyed, thresholded, shuffled bits.
13	What if a template is compromised?	Revoke the set, promote a standby.	The set becomes REVOKED; the next STANDBY set under a new key becomes ACTIVE; the leaked bits no longer match.
14	How does the key version change the template?	New version $\rightarrow$ new seeds.	The version is in the HKDF context, so projection, thresholds and shuffle all change; $\approx 50\%$ bit agreement.
15	Where is the key stored?	Nowhere; it is re-derived.	Only MASTER_SECRET lives on the server (.env); keys are recomputed with HKDF.
16	Why four template sets?	Instant replacement.	Sets 2–4 are pre-made at enrollment, so three replacements need no new capture.
17	When are the sets created?	At enrollment.	All four from the same embedding under key versions 1–4.
18	Why is no raw biometric needed to rotate?	The next templates already exist.	Rotation only changes statuses.
19	What does REVOKED mean?	Retired forever.	Never compared again; its key version is never reused.
20	What is stored in the database?	Templates + metadata.	Template bytes, key version, set number, status, timestamps, audit metadata — no images, audio, embeddings or keys.
21	What is unlinkability?	Templates under different keys can't be linked.	Cross-key agreement 0.499; $D \leftrightarrow \text{sys}$ small for face (0.018), not distinguishable from random for voice and fingerprint.
22	What if the master secret leaks?	Keys could be recreated.	It is the main trust assumption; it lives only in the server environment and a fingerprint of it is checked at startup.
23	Is the template irreversible?	Not formally claimed.	It is lossy and keyed, but no formal proof or attack study was done.
24	How can one biometric have several representations?	One embedding, several keys.	Four key versions $\rightarrow$ four unrelated templates.

Part 10 — Panel questions for Sanya

#	Question	Short answer	Detailed answer
1	How does the system decide someone is genuine?	Every presented factor must pass.	Template the live sample under the active key, compare bits, convert to an estimated metric, apply its threshold, then ALL_REQUIRED.
2	What is Hamming similarity?	Fraction of equal bits.	1.0 identical, ≈ 0.5 unrelated.
3	How do bits become a cosine?	A calibration curve.	Fitted on synthetic pairs (SYNTHETIC, $R^2 \approx 0.9999$), checked on real embeddings (error ≈ 0.07).
4	What is cosine similarity?	The angle between two vectors.	1 = same direction, 0 = unrelated; higher = more similar.
5	Why cosine for face?	The face model separates by angle.	Trained with ArcFace; embeddings have length 1.
6	What is Euclidean distance?	Straight-line distance.	For unit vectors 0–2; lower = more similar.
7	Why Euclidean for voice?	The specified voice metric.	For unit vectors $d = \sqrt{2 - 2 \cdot \cos}$: same information as cosine, stated as a distance.
8	Are 0.80 and 0.75 optimised thresholds?	No — configuration values.	Teacher-requested (TEACHER_REQUESTED_BASELINE); a development-split analysis exists but no new threshold was adopted.
9	Why different thresholds?	Different metrics and directions.	Face 0.80 (higher better), voice 0.75 (lower better).
10	How is the fingerprint matched?	Hamming ≥ 0.90 .	Directly on bits; 0.90 is a default, uncalibrated value.
11	What is fusion?	Combining factors into one decision.	evaluate_fusion_policy; ALL_REQUIRED default; WEIGHTED and AT_LEAST_TWO implemented.
12	What is fusion-scale normalisation?	One common score scale.	Everything on estimated cosine: voice via $1 - d^2/2$, fingerprint via its curve; never average a distance with a similarity.
13	Is OR a policy?	Only in the evaluation.	It was measured as a comparison rule (FRR 19.4 %, FAR 0.14 %) but is not deployable.
14	What happens when one factor fails?	Denied.	ALL_REQUIRED: other passes don't compensate.
15	What if only face is presented?	Face alone decides.	Only presented factors are evaluated.
16	What if a factor isn't enrolled?	Enrollment required (409).	Returned before any evaluation; no rotation.
17	Who decides access?	The backend.	The frontend only captures and displays; it cannot change the decision.
18	What happens after access is denied?	Audit, and rotation if enabled.	Active set retired, next standby promoted for all factors; screen shows T1 $\rightarrow$ T2.
19	What if two attacks arrive at once?	One rotation.	First takes the write lock; the second gets ALREADY_ROTATED.
20	What is POOL_EXHAUSTED?	No standby left.	Nothing changes; the user still logs in on T4.
21	Attack or noise — can you tell?	No.	The system only knows verification failed; the UI says "authentication failed".
22	What does the audit prove?	What was decided and whether it rotated.	Metadata only; no biometric data.
23	Your face FRR is 81 % — is it broken?	No, the threshold is strict.	At 0.80 false accepts are 0.0025 %; tuning on the test set would be cheating.
24	Are 657 tests 657 authentication trials?	No.	They are software tests of the code; biometric results come from the dataset evaluations.

Part 11 — Cross-questioning

Maanasvee on Reya's section

Question	Short answer
What happens to my embedding?	Four templates under four keys, then it is deleted.
Why 256 bits?	128 lost $\approx$ 7 EER points.
Can the face be rebuilt from a template?	It holds only keyed bits; no formal proof claimed.
Why four sets?	Replace instantly without re-enrollment.
Where does the key come from?	HKDF-SHA256 from the master secret + context.

Maanasvee on Sanya's section

Question	Short answer
How is my face embedding compared?	Templated under the active key; estimated cosine ≥ 0.80 passes.
Why can a genuine face fail?	Light, pose, distance; 0.80 is strict ($\approx 81\%$ false rejects on web photos).
Are there quality checks at login?	No, only at enrollment.
Cosine vs distance?	Face cosine (higher better); voice distance (lower better).
Who decides access?	The backend.

Reya on Maanasvee's section

Question	Short answer
What enters BioHash?	An L2-normalised embedding: 512 (face, fingerprint) or 192 (voice).
Which face embedding?	The centroid of at least 3 valid enrollment captures.
Which voice embedding?	The first of two recordings.
Is the embedding stored?	No — deleted after the templates are made.
Does the demo align faces?	No, bounding-box crop; alignment is off.

Reya on Sanya's section

Question	Short answer
How is the stored template used?	A live template under the same key is compared bit by bit.
What triggers rotation?	Only the backend's ACCESS_DENIED, when enabled.
What stops a double rotation?	Compare-and-swap; the second request gets ALREADY_ROTATED.
What happens after T4?	POOL_EXHAUSTED; nothing changes; user still logs in.
What is ALL_REQUIRED?	Every presented factor must pass.

Sanya on Maanasvee's section

Question	Short answer
Which models make embeddings?	InceptionResnetV1, ECAPA-TDNN, ResNet50.
Embedding sizes?	512, 192, 512.
Why can voice fail in noise?	Noise shifts the embedding, increasing distance.
How was the face model evaluated?	On 1,618 unseen LFW people (4,523 genuine / 80,900 impostor).
Was the face model trained from scratch?	No, fine-tuned from VGGFace2.

Sanya on Reya's section

Question	Short answer
Why does the user pass after rotation?	T2 came from their own enrollment, under key v2.
Is BioHash SHA-256?	No; SHA-256 only in key derivation.
What is stored?	32-byte templates + metadata.
Why is a stolen T1 useless?	Revoked and never compared; new key → unrelated template.
ACTIVE / STANDBY / REVOKED?	In use / in reserve / retired forever.

Part 12 — Difficult panel questions

Question	Accurate answer (current code)	Speaker
Why not store the raw biometric?	It is permanent and linkable; a leak can never be undone. We store only revocable templates.	Reya
Why not store embeddings?	Same problem: an embedding identifies you permanently. Templates under versioned keys can be replaced.	Reya
Why not use SHA-256 directly?	A hash changes completely when the input changes slightly; no two photos would match. SHA-256 is used only for key derivation.	Reya
Why BioHash?	It keeps similar embeddings similar (so matching works) while being keyed and replaceable.	Reya
Why 256 bits?	Measured: 128 bits lost $\approx$ 7 EER points; 256 kept most of the information.	Reya
Why four template sets?	Three instant replacements without re-enrollment; configurable pool size.	Reya
Why not make a new template from the failed biometric?	That would enroll the attacker. We only promote sets made at enrollment from the real user.	Sanya
How does the legitimate user authenticate after rotation?	Their live sample is templated under the new set's key and compared with the new set, which came from their enrollment.	Sanya
Why different thresholds?	Different metrics (similarity vs distance); both teacher-requested configuration values, not optimised.	Sanya
Why cosine for face?	ArcFace-trained, angle-based embeddings of length 1.	Sanya
Why Euclidean for voice?	Specified metric; for unit vectors it is equivalent to cosine ($d = \sqrt{2 - 2 \cdot \cos}$).	Sanya
What happens under noise?	The voice embedding shifts, the distance grows, and a genuine user can be denied — a normal verification failure.	Sanya
What happens when templates are exhausted?	POOL_EXHAUSTED: nothing changes, the user keeps working on T4, new sets need re-enrollment or authorised generate.	Sanya
What if two attackers arrive simultaneously?	Compare-and-swap under the database write lock: exactly one rotation, the other reports ALREADY_ROTATED (tested with 16 at once).	Sanya
How do you prevent mixed active sets?	One commit per rotation, one ACTIVE row per modality (unique index), one snapshot read per attempt, fail-closed security validation.	Sanya
How do you know the failed sample wasn't stored?	Rotation writes no template; a test checks the promoted set is byte-identical and no rows are added.	Sanya
How do you stop the frontend manipulating the result?	Every decision, threshold and rotation runs on the backend; the frontend only displays the response.	Sanya
Embedding vs template?	Embedding: real-valued model output (never stored). Template: 256-bit keyed BioHash output (stored).	Reya
Similarity vs distance?	Similarity: higher = more alike (face). Distance: lower = more alike (voice).	Sanya
What is cancellability?	Issuing a new, unrelated protected template from the same biometric by changing the key.	Reya
What is revocability?	Retiring a template set permanently so it is never compared again.	Reya
Your face false-reject rate is 81 % — is it broken?	No: 0.80 is very strict on web photos; false accepts are 0.0025 %. Tuning it on the test set would be cheating.	Sanya
Does the demo use face alignment?	No, bounding-box crop; alignment is implemented and measured but off.	Maanasvee
Did you train the face model from scratch?	No: VGGFace2-pretrained, fine-tuned on 62 LFW identities; voice was trained from scratch.	Maanasvee
Isn't 24 speakers too few?	Yes, it is a stated limitation; speakers are shared across splits.	Maanasvee
Can someone lock a user out by failing on purpose?	No lock-out: after three failures rotation stops (POOL_EXHAUSTED). But standby sets are spent, so rotation is off by default.	Sanya

Part 13 — What to actually say: Maanasvee

1. Datasets

What to say: We used LFW for faces, a 24-speaker VoxCeleb1 subset for voice, and SOCOFing for fingerprints — free, labelled research datasets.

Key point: Evaluation used 1,618 people the face model never saw.

Technical detail: Face fine-tuning: 62 people / 3,023 images.

Likely question: Why these? **Answer:** Free, standard, small enough for our hardware.

2. Training

What to say: The face model was pretrained on VGGFace2 and we fine-tuned only its last layers. Fingerprint started from ImageNet. Voice was trained from scratch.

Key point: Only voice was trained from scratch.

Technical detail: All used ArcFace loss.

Likely question: Did you train the face model? **Answer:** Fine-tuned, not from scratch.

3. Models

What to say: InceptionResnetV1 for face, ECAPA-TDNN for voice, ResNet50 for fingerprint.

Key point: Three networks, one per factor.

Technical detail: MTCNN is the face detector, not the recogniser.

Likely question: Which model makes the face embedding? **Answer:** InceptionResnetV1.

4. Embeddings

What to say: Each model outputs an embedding — 512 numbers for face and fingerprint, 192 for voice — scaled to length 1.

Key point: Same person → nearby points.

Technical detail: Two captures differ slightly but stay close.

Likely question: Why not compare pixels? **Answer:** Pixels change with light and pose.

5. Face processing

What to say: MTCNN finds the face and five landmarks; we crop the face box to 160 by 160 and the network produces the embedding.

Key point: Deployed crop = bounding box.

Technical detail: Pixel scaling $(x - 127.5)/128$ matches training.

Likely question: What input size? **Answer:** 160×160×3.

6. Face alignment

What to say: Alignment straightens the face using the landmarks. We implemented and measured it — with a retrained model it cut the error rate from 8.67 to 7.18 percent — but the demo still uses the plain crop, because switching needs the new model and everyone to re-enroll.

Key point: Implemented and measured, not deployed.

Technical detail: Old model + aligned input got worse.

Likely question: So why not switch? **Answer:** It needs re-enrollment of all users.

7. Five-capture enrollment

What to say: Each of the five captures is checked immediately and nothing is stored. After five good ones, the server re-checks them, averages at least three into one embedding, and deletes the rest.

Key point: ≥ 3 valid → one centroid.

Technical detail: Gates: confidence, sharpness, size, centring, tilt, turn.

Likely question: What if one fails? **Answer:** Retake; the counter doesn't move.

8. Voice processing

What to say: Resample to 16 kHz, trim silence, normalise loudness, take 4 seconds, compute an 80-band log-mel spectrogram, and ECAPA-TDNN gives 192 numbers. Two enrollment recordings must agree.

Key point: Templates come from the first recording.

Technical detail: Consistency ≥ 0.75 enrolled; 0.60–0.74 warning.

Likely question: Why two recordings? **Answer:** To check consistency.

9. Fingerprint processing

What to say: CLAHE and Gabor filters sharpen the ridges, the image is resized to 224 by 224, and ResNet50 gives 512 numbers.

Key point: Fingerprint is optional, single image.

Technical detail: EER 30.77 % — the weakest model.

Likely question: Is there a scanner? **Answer:** No, images are uploaded.

10. Handoff to Reya

What to say: Every embedding goes straight into the protection layer in memory and is never saved. Reya will now explain how that embedding becomes a protected, revocable template.

Key point: Embeddings are never stored.

Technical detail: Only 32-byte templates reach the database.

Likely question: Where does my face end up? **Answer:** As four protected 256-bit templates.

Part 13 — What to actually say: Reya

1. The problem

What to say: Passwords can be changed after a leak; faces, voices and fingerprints can't. If a system stores your biometric template and it leaks, it's compromised for life — and it can link you across services.

Key point: Biometrics are permanent, so stored templates must be replaceable.

Technical detail: Traditional systems store the embedding itself.

Likely question: What are you protecting? **Answer:** The stored biometric representation.

2. Cancellable biometrics

What to say: So we never store the biometric representation. We store a version transformed with a secret key, and if it leaks we switch keys and issue a new, unrelated template.

Key point: Same face, new key, new template.

Technical detail: Old versions are retired and never reused.

Likely question: How can a face be replaceable? **Answer:** The face isn't — its protected representation is.

3. Keys

What to say: The keys come from HKDF-SHA256: the server's master secret plus the application, the user, the factor and a version number. Change any of them and you get a completely different template.

Key point: Version = which key.

Technical detail: Salt = SHA-256 of the context; three seeds.

Likely question: Where is the key stored? **Answer:** Nowhere — it's re-derived.

4. BioHash

What to say: Maanasvee's layer gives us an embedding. BioHash scales it to length 1, projects it onto 256 secret orthonormal directions, turns each value into a bit with a secret threshold, and shuffles the bits.

Key point: BioHash keeps similar faces similar, unlike a hash.

Technical detail: QR-orthonormal projection; thresholds with spread $0.5 \times$ data spread; keyed permutation.

Likely question: Is it SHA-256? **Answer:** No — SHA-256 is only used to derive the keys.

5. The 256-bit template

What to say: The result is 256 bits, 32 bytes. That is all we store — no image, audio or embedding.

Key point: 256 bits = 32 bytes.

Technical detail: 128 bits lost about 7 EER points in our experiment.

Likely question: Why 256? **Answer:** We measured it.

6. Four template sets

What to say: At enrollment we make four templates per factor under four versions: T1 is active, T2 to T4 wait on standby. Each set holds every factor the user enrolled.

Key point: Four sets = three instant replacements.

Technical detail: Only enrolled factors have rows.

Likely question: When are they made? **Answer:** All at enrollment.

7. ACTIVE / STANDBY / REVOKED

What to say: Only the active set is ever compared. Standby sets wait. Revoked sets are retired forever.

Key point: Only ACTIVE is compared.

Technical detail: A unique index allows one active row per factor.

Likely question: Can a revoked set come back? **Answer:** No, its key version is never reused.

8. Storage and security

What to say: The database holds template bytes, key versions, set numbers, statuses and audit metadata. No images, no audio, no embeddings, no keys. Templates under different keys agree on only about half their bits, so they can't be linked.

Key point: A database thief gets only revocable templates.

Technical detail: Cross-key agreement 0.499; face D↔sys small (0.018).

Likely question: What if the database leaks? **Answer:** Revoke the sets.

9. Revocation → handoff

What to say: If T1 is compromised it becomes revoked and T2 becomes active. The user logs in normally because T2 came from their own enrollment — no new capture needed. Sanya will now show how a login uses the active set, and what happens when one fails.

Key point: Revocation needs no raw biometric.

Technical detail: It only changes statuses.

Likely question: Why does the user still pass? **Answer:** Their live sample is templated under T2's key.

Part 13 — What to actually say: Sanya

1. Authentication starts

What to say: The user picks the factors they enrolled and captures them. The backend checks they're all enrolled, then reads the user's template sets once for the whole attempt.

Key point: The backend decides; the browser just displays.

Technical detail: Unenrolled factor → 409 ENROLLMENT_REQUIRED.

Likely question: Can the frontend fake a pass? **Answer:** No, every decision is on the server.

2. Matching

What to say: We template the new sample with the active key and compare it with the stored template bit by bit; a calibration curve turns the fraction of equal bits into an estimated cosine or distance.

Key point: We compare templates, never embeddings.

Technical detail: Curve fit is SYNTHETIC ($R^2 \approx 0.9999$); real genuine-pair error ≈ 0.07 .

Likely question: Is it a real cosine? **Answer:** A calibrated estimate.

3. Face

What to say: The face estimate must be 0.80 or higher.

Key point: Face uses estimated cosine ≥ 0.80 .

Technical detail: ≈ 210 of 256 bits must match.

Likely question: Why cosine? **Answer:** The model separates people by angle.

4. Voice

What to say: Voice uses a distance, so lower is better: 0.75 or less passes.

Key point: Voice: distance ≤ 0.75 .

Technical detail: Equals estimated cosine ≥ 0.719 .

Likely question: Why can voice fail? **Answer:** Noise shifts the embedding.

5. Fingerprint

What to say: Fingerprint is compared directly on bits: at least 90 percent must match.

Key point: Fingerprint: Hamming ≥ 0.90 .

Technical detail: A default, uncalibrated value.

Likely question: Is fingerprint required? **Answer:** No, it is optional.

6. Thresholds

What to say: Face 0.80 and voice 0.75 are the values our teacher asked for — configuration values, not thresholds we optimised.

Key point: Thresholds are configuration, not tuned.

Technical detail: Face at 0.80: 81.5 % false rejects, 0.0025 % false accepts.

Likely question: Why not lower them? **Answer:** Choosing a threshold on the test set would be cheating; it needs real users.

7. Fusion

What to say: Every score is put on one scale, and our default policy is ALL_REQUIRED: every factor presented must pass. WEIGHTED and AT_LEAST_TWO are also implemented; OR was only evaluated.

Key point: ALL_REQUIRED over presented factors.

Technical detail: Voice normalised as $1 - d^2/2$.

Likely question: Face fails, voice passes? **Answer:** Denied.

8. Access decision

What to say: If everything passes, access is granted and nothing rotates; otherwise it is denied.

Key point: Granted → no rotation.

Technical detail: Denied responses never reveal the user's name.

Likely question: What is returned? **Answer:** Decision, fused score, set version.

9. Failed authentication and rotation

What to say: The backend knows verification failed — not whether it was an attacker or a noisy microphone. With rotation on, a denial retires T1 and promotes T2 for every factor at once, and nothing is made from the failed sample.

Key point: Never claim "attack detected".

Technical detail: `crud.rotate_active_set_if_current`, one commit.

Likely question: Where did T2 come from? **Answer:** Enrollment, key v2.

10. Concurrency

What to say: If two failures arrive together, compare-and-swap retires T1 only if it's still active; the second request gets `ALREADY_ROTATED`.

Key point: Exactly one rotation per set.

Technical detail: Write lock + row-count check; tested with 16 at once.

Likely question: What if the pool runs out? **Answer:** `POOL_EXHAUSTED`; the user still logs in.

11. Audit and results (close)

What to say: Every attempt writes one audit row of metadata — never images or embeddings. On real data the protected templates reach 8.50 % EER for face and 2.32 % for voice. Raw biometric, embedding, protected template, decision, rotation — that's our system.

Key point: Results are REAL DATA; the 657 tests are software tests.

Technical detail: Fusion results use chimeric users.

Likely question: Are 657 tests 657 trials? **Answer:** No — they test the software.

Part 14 — Transitions

Maanasvee → Reya

Say: "So every face, voice and fingerprint ends up as an embedding — 512 or 192 numbers that describe who you are — and that embedding is never saved. But an embedding is permanent: if it leaked, it would identify you forever. Reya will now explain how we turn it into a protected template that can be revoked and replaced."

Reya → Sanya

Say: "So every biometric is stored only as protected 256-bit templates, in four sets with different keys, and only one set is ever active. We never store an image, a recording or an embedding. Sanya will now show how the system uses that active set when someone tries to log in, how the final decision is made, and what happens when verification fails."

Sanya — final system summary

Say: "To close the loop: Maanasvee showed how a live sample becomes an embedding; Reya showed how it becomes four protected templates. At login, our server compares a fresh template with the active set, applies the configured thresholds and ALL_REQUIRED, grants or denies access, and can rotate to the next set after a denial. Raw biometric, embedding, protected template, decision, rotation — that's our system. Thank you."

Part 15 — The 2-minute version and 30-second summaries

The whole project in 2 minutes

"Our project is a biometric login for critical infrastructure that never stores your biometric. **First**, a camera frame, voice recording or fingerprint image reaches our server. For face, MTCNN finds the face and we crop it; for voice we clean the audio into a spectrogram; for fingerprint we enhance the ridges. Trained networks — InceptionResnetV1, ECAPA-TDNN and ResNet50 — turn each into an embedding: 512 or 192 numbers that describe who it is. **Second**, we never store that embedding. BioHash projects it onto 256 secret directions, turns each into a bit and shuffles them, using a key derived from a server secret plus the user, the factor and a version number. At enrollment we make four such templates: T1 is active, T2 to T4 wait in reserve. The database holds only these 32-byte templates. **Third**, at login we template the live sample under the active key and count matching bits. Face must reach an estimated cosine of 0.80, voice an estimated distance of 0.75 or less — both configuration values set by our teacher — fingerprint 90 % matching bits, and under ALL_REQUIRED every factor presented must pass. If access is denied and rotation is on, T1 is retired and T2 becomes active for all factors at once — safely, even when two failures arrive together, using a compare-and-swap. The real user keeps logging in on T2 because it came from their own enrollment, and the failed sample is never stored."

Maanasvee — 30 seconds

Say: "We used LFW, VoxCeleb and SOCOFing. The face model is a fine-tuned VGGFace2 network, the voice model was trained from scratch, and fingerprint is a fine-tuned ResNet50. MTCNN finds the face, five captures are checked and averaged, and each factor becomes a 512- or 192-number embedding that goes straight into the protection layer. Alignment is implemented and measured but not yet deployed."

Reya — 30 seconds

Say: "Biometrics can't be changed like passwords, so we never store them. Each embedding goes through BioHash — a keyed orthonormal projection, threshold and shuffle, with keys from HKDF-SHA256 — into a 256-bit template. At enrollment we make four templates under four key versions; only one set is active. If it's compromised we revoke it and promote the next, and the user keeps working with no new capture."

Sanya — 30 seconds

Say: "At login the server templates the live sample under the active key and compares bits. Face needs an estimated cosine of 0.80, voice a distance of 0.75 or less — configuration values, not tuned — fingerprint 90 % matching bits, and ALL_REQUIRED needs every presented factor to pass. After a denial, the server can rotate every factor to the next set in one safe compare-and-swap, without storing the failed sample."

Part 16 — Last-minute cheat sheet

Speaker	Owns
Maanasvee	datasets · training · preprocessing · quality gates · alignment status · embeddings
Reya	cancellable biometrics · HKDF · BioHash · 256-bit template · four sets · revocation · unlinkability
Sanya	authentication · Hamming → estimates · thresholds · fusion · rotation · concurrency · audit · results

Factor	Metric	Direction	Threshold (configuration)	Model · embedding
Face	estimated cosine	higher = better	≥ 0.80	InceptionResnetV1 · 512
Voice	estimated Euclidean distance	lower = better	≤ 0.75	ECAPA-TDNN · 192
Fingerprint	Hamming similarity	higher = better	≥ 0.90	ResNet50 · 512

Term	One-line meaning
Embedding	numbers a network outputs to describe identity (never stored)
Template	256-bit BioHash output = 32 bytes (stored)
BioHash	L2 → keyed orthonormal projection → keyed quantisation → keyed permutation
HKDF-SHA256	derives the three seeds from the master secret + app user modality version
Template sets	T1 ACTIVE, T2–T4 STANDBY, all made at enrollment
ACTIVE / STANDBY / REVOKED	compared now / in reserve / retired forever
Hamming similarity	fraction of equal bits (1 = identical, ≈ 0.5 unrelated)
ALL_REQUIRED	every presented factor must pass (default)
WEIGHTED / AT_LEAST_TWO	implemented alternatives; OR = evaluation-only
ENROLLMENT_REQUIRED	presented factor not enrolled → 409, nothing evaluated
Rotation	on DENIED (if enabled): T _n → REVOKED, T _{n+1} → ACTIVE, all factors, one commit
Compare-and-swap	retire the set only if still ACTIVE; loser gets ALREADY_ROTATED
POOL_EXHAUSTED	no standby left; nothing changes; user still logs in
Cancellability	new key → new unrelated template from the same biometric
Revocability	retire a set permanently

Flows to remember

Enrollment: capture → preprocess → embedding → BioHash ×4 keys → T1 ACTIVE, T2–T4 STANDBY → embedding deleted
 Login: capture → embedding → template (active key) → Hamming → estimate → threshold → ALL_REQUIRED
 Denied: (rotation on) → compare-and-swap → T_{n+1} ACTIVE → audit → result page
 Face: MTCNN → bbox crop 160×160 (alignment off) → InceptionResnetV1 → 512 → L2
 Voice: 16 kHz → trim → RMS → 4 s → 80 log-mel → ECAPA-TDNN → 192 → L2
 Fingerprint: CLAHE → ridge norm → Gabor ×8 → 224×224 → ResNet50 → 512 → L2

Differences examiners like to test

	A	B
Embedding vs template	real numbers from a network, never stored	256 keyed bits, stored
Similarity vs distance	higher = more alike (face)	lower = more alike (voice)
Hash vs BioHash	tiny change → totally different output	similar inputs stay similar
Revoke vs re-enroll	switch to a pre-made set from the same enrollment	capture the biometric again
Configuration vs optimised threshold	0.80 / 0.75 set by the teacher	not done (needs real users)
Synthetic calibration vs real results	curve fit on generated vectors	EER measured on real datasets
Software tests vs trials	657 code tests	not biometric trials
Alignment implemented vs deployed	implemented and measured (8.67 → 7.18 % EER)	deployed = bounding-box crop

Common questions — one-line answers

- **Did you train the face model?** Fine-tuned VGGFace2 model; voice from scratch. (Maanasvee)
- **Does the demo align faces?** No, bounding-box crop. (Maanasvee)
- **Why five captures?** ≥ 3 valid are averaged into a steadier embedding. (Maanasvee)
- **Is BioHash SHA-256?** No; SHA-256 is only in key derivation. (Reya)
- **Why 256 bits?** 128 lost $\approx$ 7 EER points. (Reya)
- **Why four sets?** Three instant replacements, no re-enrollment. (Reya)
- **What's stored?** 32-byte templates + metadata; no images, audio, embeddings, keys. (Reya)
- **Who decides access?** The backend. (Sanya)
- **Are 0.80 / 0.75 optimised?** No, teacher-requested configuration. (Sanya)
- **Face fails, voice passes?** Denied (ALL_REQUIRED). (Sanya)
- **Two failures at once?** One rotation; the other is ALREADY_ROTATED. (Sanya)
- **After T4?** POOL_EXHAUSTED; user still logs in. (Sanya)
- **Is rotation always on?** No, off by default; on for the demo. (Sanya)